Toegepaste Informatica

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Front-End Development

Part 2: Dynamic web applications

JS: DOM manipulation & Event handling

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JS: DOM

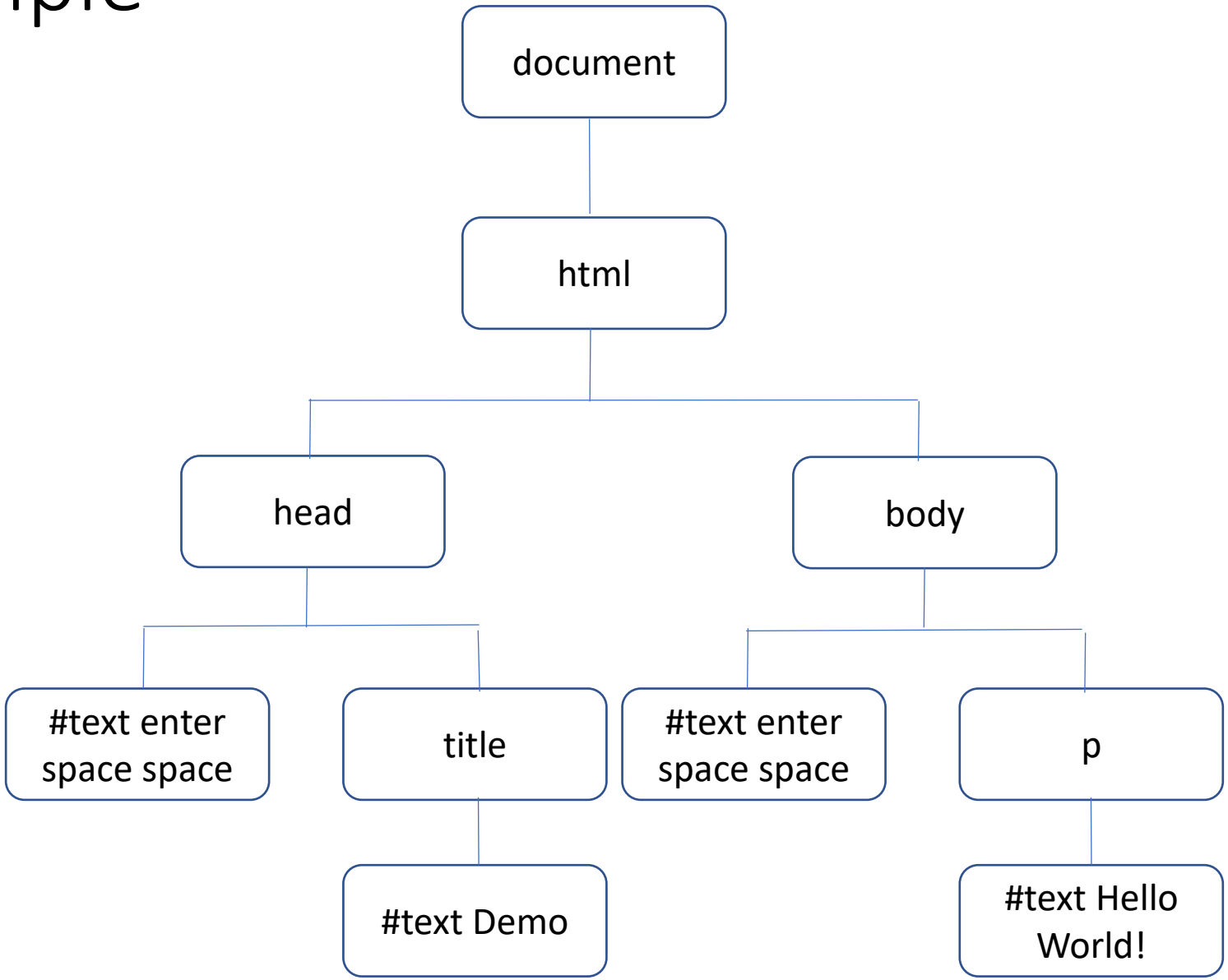
- DOM Manipulation
  - DOM tree
  - Finding elements in DOM tree
    - getElementById
    - querySelector
  - Traversing DOM tree
  - Adding &changing attributes
  - Creating & adding elements
- Events
  - Event listener

DOM tree

- DOM = Document Object Model
  - Every HTML element is an object
- DOM Tree
  - The HTML elements of a particular page are organised as a tree with parent nodes and children
    - Element nodes (HTML)
    - Text nodes
    - Attribute nodes

DOM tree - Example

```html
<!DOCTYPE html>
<html>
  <head>
    <title>Demo</title>
  </head>
  <body>
    <p>Hello World!</p>
  </body>
</html>
```



DOM tree in Web Developer Tool

Hello World!

Inspector | Console | Debugger | Network | Style Editor | Performance | Memory | Storage | Accessibility | Application

Search HTML

```
<!DOCTYPE html>
<html>
  <head>
    <title>Demo</title>
  </head>
  <body>
    <p style="background: red;">Hello World!</p>
  </body>
</html>
```

html > body > p

Filter Styles :hov .cls

```
element {                                inline
    background: red;
}
```

Layout | Computed | Changes | Compatibility

Flexbox

Select a Flex container or item to continue.

Grid

CSS Grid is not in use on this page

Box Model

margin 16
border 0
padding 0
0 0 0 1105×19.2 0 0 0

Filter Output | Errors | Warnings | Logs | Info | Debug | CSS | XHR | Requests

```
» temp0.style.background='red'
← "red"
»
```

Finding elements in DOM tree

- getElementById
  - returns an Element object representing the element whose id property matches the specified string
  - since element IDs are required to be unique if specified, they're a useful way to get access to a specific element quickly

```html
<form id="add-animal">
  <p><label for="name">Name:</label><input id="name" type="text" /></p>
  <p><input id="submit" type="submit" value="Add" /></p>
</form>
```

```javascript
document.getElementById("add-animal")
```

Finding elements in DOM tree

- querySelector
  - returns the first Element within the document that matches the specified (CSS) selector, or group of selectors
  - if no matches are found, null is returned

```
<form id="add-animal">
  <p><label for="name">Name:</label><input id="name" type="text" /></p>
  <p><input id="submit" type="submit" value="Add" /></p>
</form>
```

```
document.querySelector('form')
```

Finding Elements in DOM tree

- https://developer.mozilla.org/en-US/docs/Web/API/Document/getElementById
- https://developer.mozilla.org/en-US/docs/Web/API/Document/getElementsByTagName
- https://developer.mozilla.org/en-US/docs/Web/API/Document/getElementsByClassName
- https://developer.mozilla.org/en-US/docs/Web/API/Document/querySelector
- https://developer.mozilla.org/en-US/docs/Web/API/Document/querySelectorAll

Traversing DOM tree

- Some methods to traverse throughout the DOM tree
  - parentNode
  - childNodes[*nodenumber*]
  - firstChild, lastChild
  - nextSibling, previousSibling

```html
<form id="add-animal">
  <p><label for="name">Name:</label><input id="name" type="text" /></p>
  <p><input id="submit" type="submit" value="Add" /></p>
</form>
```

```javascript
// value of the input field with id name
// by id ...
// const name = document.getElementById("name").value
// by children ...
const name = document.getElementById("add-animal").children[0].children[1].value
```

Traversing DOM tree

- https://developer.mozilla.org/en-US/docs/Web/API/Element/children
- https://developer.mozilla.org/en-US/docs/Web/API/Node/firstChild
- https://developer.mozilla.org/en-US/docs/Web/API/Node/parentElement

Adding & changing attributes

- Sets the value of an attribute on the specified element. If the attribute already exists, the value is updated; otherwise a new attribute is added with the specified name and value.
  - setAttribute(name, value)
  - element.attribute = value

```html
<form id="add-animal">
  <p><label for="name">Name:</label><input id="name" type="text" /></p>
  <p><input id="submit" type="submit" value="Add" /></p>
</form>
```

```javascript
document.getElementById("name").type="button"
document.getElementById("name").setAttribute("style", "background-color: green")

// OR

document.getElementById("name").style.background-color ="green"
document.getElementById("name").className="active"
```

Adding and changing attributes

- https://developer.mozilla.org/en-US/docs/Web/API/Element/setAttribute
- https://developer.mozilla.org/en-US/docs/Web/API/Element/removeAttribute

Creating & adding elements

- Some methods for creating and adding elements
  - createElement
  - appendChild

```html
<main>
  <h2>Animals currently on The Farm</h2>
  <div id="animals"></div>
</main>
```

```javascript
const p = document.createElement("p")
p.innerHTML = "something"
document.getElementById("animals").appendChild(p
```

Events

- Events are actions or occurrences that happen in the system you are programming, which the system tells you about so your code can react to them.
  - The user selects, clicks, or hovers the cursor over a certain element
  - The user chooses a key on the keyboard
  - A web page finishes loading
  - A form is submitted
  - …

Event listener

- addEventListener()
  - sets up a function that will be called whenever the specified event is delivered to the target
  - common targets are element, or its children, document, and window, but the target may be any object that supports events
- there also exists a function removeEventListener()

```html
<form id="add-animal">
  <p><input id="submit" type="submit" value="Add" /></p>
</form>
```

```javascript
document.getElementById('add-animal').addEventListener('submit', (event) => {
  event.preventDefault();
  handleAddAnimal();
});
```

Event listener

- https://developer.mozilla.org/en-US/docs/Web/Events#event_listing
  e.g. mouseover, mouseout, click, blur, focus, input

References

- https://developer.mozilla.org/en-US/docs/Learn/JavaScript
- https://www.freecodecamp.org/news/manage-default-behavior-in-browser/